

CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge International Advanced Subsidiary and Advanced Level

MARK SCHEME for the October/November 2014 series

9702 PHYSICS

9702/21

Paper 2 (AS Structured Questions), maximum raw mark 60

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the October/November 2014 series for most Cambridge IGCSE[®], Cambridge International A and AS Level components and some Cambridge O Level components.

® IGCSE is the registered trademark of Cambridge International Examinations.

Page 2	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2014	9702	21

- 1 (a) temperature
current
(allow amount of substance and luminous intensity) B1 B1 [2]
- (b) base units of force constant: $\text{kg m s}^{-2} \text{m}^{-1}$ or kg s^{-2} B1
base units of time and mass: s and kg C1
base units of C: $\text{s (kg s}^{-2} / \text{kg)}^{1/2}$ cancelling to show no units B1 [3]
- 2 (a) pressure = force / area (normal to the force) [clear ratio essential] B1 [1]
- (b) (i) $P = mg / A = (5.09 \times 9.81) / A$ C1
 $A = (\pi d^2 / 4) = \pi \times (9.4 \times 10^{-2})^2 / 4 (= 0.00694 \text{ m}^2)$ C1
 $P = 49.93 / 0.00694$
 $= 7200 \text{ (7195) Pa}$ (minimum of 2 s.f. required) A1 [3]
- (ii) $\Delta P / P = \Delta m / m + 2\Delta d / d$ C1
 $= 0.01 / 5.09 + (2 \times 0.1) / 9.4 (= 0.0020 + 0.021 \text{ or } 2.3\%)$ C1
 $\Delta P = 170 \text{ (165 to 167) Pa}$ A1 [3]
- (iii) $P = 7200 \pm 200 \text{ Pa}$ A1 [1]
- 3 (a) random error (in the measurements) of the length OR resistance B1 [1]
- (b) gradient = $(3.6 - 1.9) / (0.8 - 0.4)$ C1
 $= 4.25$ A1 [2]
- (c) $R = \rho l / A$ C1
 $\rho = \text{gradient} \times \text{area} = 4.25 \times 0.12 \times 10^{-6}$ C1
 $= 5.1(0) \times 10^{-7} \Omega \text{ m}$ A1 [3]
- (d) resistance decreasing with increasing area B1
correct shape with curve being asymptote to both axes B1 [2]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2014	9702	21

- 4 (a) (i) acceleration = $(v - u) / t$ or $(12 - 0.5) / 4$ C1
 $= (12 - 0.5) / 4 = 2.9$ (2.875) (= approximately 3 ms^{-2}) M1 [2]
- (ii) $x = (u + v)t / 2$
 $= [(12 + 0.5) \times 4] / 2$ C1
 $= 25 \text{ m}$ A1 [2]
- (iii) line with increasing gradient M1
non-zero gradient at origin A1 [2]
- (b) (i) weight down slope = $2 \times 9.81 \times \sin 25^\circ = 8.29 / 8.3$ M1 [1]
- (ii) ($F = ma$) $8.3 - F_R = 2 \times 2.9$ C1
 $F_R = 2.5$ (2.3 if 3 used for a) N A1 [2]
- 5 (a) (i) change in kinetic energy = $\frac{1}{2}mv^2$ C1
 $= 0.5 \times 25 \times (0.64)^2 = 5.1(2) \text{ J}$ A1 [2]
- (ii) zero A1 [1]
- (iii) $(-)5.1(2) \text{ J}$ A1 [1]
- (b) (i) $PE = mgh$ C1
 $= 350 \times 0.64 \times 25$ C1
 $= 5600 \text{ J}$ A1 [3]
(If full length used allow 1/3)
- (ii) $P = Fv$ or gain in PE / t , E_P / t or work done $/ t$, W / t C1
 $= 350 \times 0.64$ or $5600 / 25$
 $= 220$ (224) W A1 [2]
- 6 melting: solid to liquid B1
at a specific/one temperature/at the melting point B1
- evaporation: liquid to vapour/gas OR molecules escape from surface of liquid B1
at all temperatures B1 [4]

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2014	9702	21

- 7 (a) due to the lost volts in internal resistance / cell or energy losses in the internal resistance / cell B1 [1]
- (b) (i) $V = IR$ C1
 $= 1.2 \times 6 = 7.2 \text{ V}$ A1 [2]
- (ii) p.d. across Y and internal resistance $r = 4.8 \text{ (V)}$ [12 – 7.2] C1
resistance of Y + $r = 4.8 / 1.2 = 4 \text{ (}\Omega\text{)}$ C1
resistance of Y = $4 - 0.5 = 3.5 \text{ }\Omega$ A1 [3]
or
 $R_{\text{total}} = 12 / 1.2 = 10 \text{ (}\Omega\text{)}$ (C1)
 $X + r = 6.5 \text{ (}\Omega\text{)}$ (C1)
resistance of Y = $3.5 \text{ }\Omega$ (A1)
- (iii) $P = I^2 r$ C1
 $= (1.2)^2 \times 0.5 = 0.72 \text{ W}$ A1 [2]
- (c) terminal p.d. increases as R is increased B1 [1]
current decreases so there are less lost volts
- 8 (a) two waves (of the same kind) travelling in opposite directions overlap B1
waves have same frequency / wavelength and speed B1 [2]
- (b) (i) $T = 0.8 \text{ (ms)}$ C1
 $f = 1 / (0.8 \times 10^{-3}) = 1250 \text{ (Hz)}$ A1 [2]
- (ii) microphone is moved from plate to loudspeaker or vice versa B1
wavelength is the twice the distance between adjacent maxima or minima (seen on c.r.o.) B1 [2]
- (iii) $v = f\lambda$ C1
 $= 1250 \times 0.26$
 $= 330 \text{ (325) m s}^{-1}$ A1 [2]